```
Clear["Global`*"];

Import["path/SiO2.txt"];

Si = ImportString[%2]
```

```
{{0.43, 1.46721}, {0.441, 1.46628}, {0.452, 1.46542},
 {0.463, 1.46463}, {0.474, 1.4639}, {0.485, 1.4632}, {0.496, 1.46256},
 {0.507, 1.46196}, {0.518, 1.46138}, {0.529, 1.46086},
 {0.54, 1.46035}, {0.551, 1.45987}, {0.562, 1.45943}, {0.573, 1.45899},
 {0.584, 1.4586}, {0.595, 1.45821}, {0.606, 1.45785}, {0.617, 1.45749},
 {0.628, 1.45717}, {0.639, 1.45684}, {0.65, 1.45654}, {0.661, 1.45624},
 {0.672, 1.45597}, {0.683, 1.45569}, {0.694, 1.45544},
 {0.705, 1.45518}, {0.716, 1.45494}, {0.727, 1.45471}, {0.738, 1.45448},
 {0.749, 1.45426}, {0.76, 1.45405}, {0.771, 1.45384}, {0.782, 1.45364},
 {0.793, 1.45344}, {0.804, 1.45325}, {0.815, 1.45306}, {0.826, 1.45288},
 {0.837, 1.4527}, {0.848, 1.45253}, {0.859, 1.45236}, {0.87, 1.45219},
 {0.881, 1.45203}, {0.892, 1.45187}, {0.903, 1.45171}, {0.914, 1.45156},
 {0.925, 1.45141}, {0.936, 1.45125}, {0.947, 1.45111}, {0.958, 1.45096},
 {0.969, 1.45082}, {0.98, 1.45067}, {0.991, 1.45053}, {1.002, 1.45039},
 {1.013, 1.45026}, {1.024, 1.45012}, {1.035, 1.44998}, {1.046, 1.44985},
 {1.057, 1.44972}, {1.068, 1.44958}, {1.079, 1.44945}, {1.09, 1.44932},
 {1.101, 1.44919}, {1.112, 1.44906}, {1.123, 1.44894}, {1.134, 1.44881},
 {1.145, 1.44868}, {1.156, 1.44855}, {1.167, 1.44843}, {1.178, 1.4483},
 {1.189, 1.44818}, {1.2, 1.44805}, {1.211, 1.44793}, {1.222, 1.4478},
 {1.233, 1.44768}, {1.244, 1.44755}, {1.255, 1.44743}, {1.266, 1.4473},
 {1.277, 1.44718}, {1.288, 1.44705}, {1.299, 1.44693}, {1.31, 1.4468},
 {1.321, 1.44668}, {1.332, 1.44655}, {1.343, 1.44643}, {1.354, 1.4463},
 {1.365, 1.44618}, {1.376, 1.44605}, {1.387, 1.44593}, {1.398, 1.4458},
 {1.409, 1.44568}, {1.42, 1.44555}, {1.431, 1.44542}, {1.442, 1.44529},
 {1.453, 1.44517}, {1.464, 1.44504}, {1.475, 1.44491}, {1.486, 1.44478},
 {1.497, 1.44465}, {1.508, 1.44452}, {1.519, 1.44439}, {1.53, 1.44426}}
```

```
Import["path/TiO2.txt"];
```

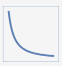
```
Ti = ImportString[%4]
```

```
{{0.43, 2.8717}, {0.441, 2.8374}, {0.452, 2.8074},
 {0.463, 2.78095}, {0.474, 2.75749}, {0.485, 2.73653},
 {0.496, 2.71772}, {0.507, 2.70075}, {0.518, 2.68537},
 {0.529, 2.67139}, {0.54, 2.65861}, {0.551, 2.64691}, {0.562, 2.63616},
 {0.573, 2.62624}, {0.584, 2.61708}, {0.595, 2.6086}, {0.606, 2.60071},
 {0.617, 2.59338}, {0.628, 2.58654}, {0.639, 2.58015}, {0.65, 2.57417},
 {0.661, 2.56856}, {0.672, 2.56329}, {0.683, 2.55833}, {0.694, 2.55367},
 {0.705, 2.54927}, {0.716, 2.54511}, {0.727, 2.54118}, {0.738, 2.53746},
 {0.749, 2.53393}, {0.76, 2.53058}, {0.771, 2.5274}, {0.782, 2.52438},
 {0.793, 2.52151}, {0.804, 2.51877}, {0.815, 2.51615}, {0.826, 2.51366},
 {0.837, 2.51128}, {0.848, 2.50901}, {0.859, 2.50683}, {0.87, 2.50475},
 {0.881, 2.50276}, {0.892, 2.50084}, {0.903, 2.49901}, {0.914, 2.49725},
 {0.925, 2.49556}, {0.936, 2.49394}, {0.947, 2.49238}, {0.958, 2.49088},
 {0.969, 2.48943}, {0.98, 2.48804}, {0.991, 2.4867}, {1.002, 2.48541},
 {1.013, 2.48416}, {1.024, 2.48296}, {1.035, 2.4818}, {1.046, 2.48068},
 {1.057, 2.4796}, {1.068, 2.47855}, {1.079, 2.47754}, {1.09, 2.47656},
 {1.101, 2.47561}, {1.112, 2.47469}, {1.123, 2.47381}, {1.134, 2.47295},
 {1.145, 2.47211}, {1.156, 2.4713}, {1.167, 2.47052}, {1.178, 2.46976},
 {1.189, 2.46902}, {1.2, 2.4683}, {1.211, 2.46761}, {1.222, 2.46693},
 {1.233, 2.46628}, {1.244, 2.46564}, {1.255, 2.46502}, {1.266, 2.46441},
 {1.277, 2.46383}, {1.288, 2.46326}, {1.299, 2.4627}, {1.31, 2.46216},
 {1.321, 2.46163}, {1.332, 2.46112}, {1.343, 2.46062}, {1.354, 2.46013},
 {1.365, 2.45966}, {1.376, 2.45919}, {1.387, 2.45874}, {1.398, 2.4583},
 {1.409, 2.45787}, {1.42, 2.45745}, {1.431, 2.45704}, {1.442, 2.45665},
 {1.453, 2.45626}, {1.464, 2.45588}, {1.475, 2.4555}, {1.486, 2.45514},
 {1.497, 2.45479}, {1.508, 2.45444}, {1.519, 2.4541}, {1.53, 2.45377}}
```

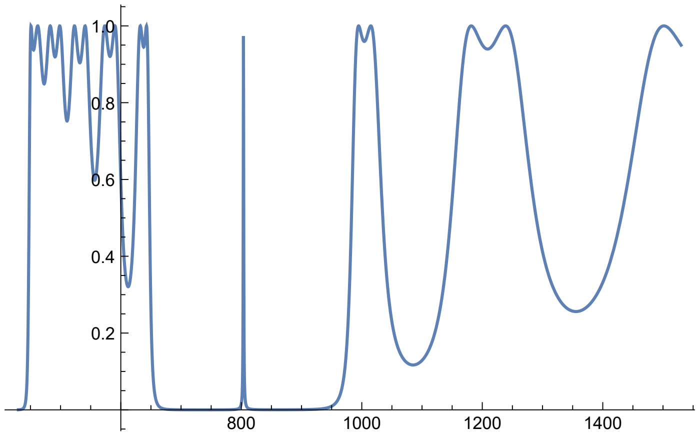
```
n1 = Interpolation[Si]
```

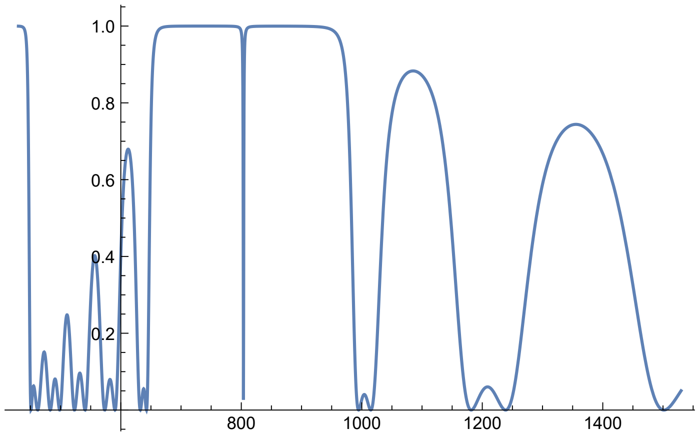
InterpolatingFunction[ Domain: {{0.43, 1.53}} Output: scalar]

```
n2 = Interpolation[Ti]
```

InterpolatingFunction[ Domain: {{0.43, 1.53}} Output: scalar]

```
h1 = 110;
h2 = 90;
hd = 300;
Nbef = 6;
Naft = 6;
n0 = 1;
n01 = 1;
ϵ = 1;
μ = 1;
θ = 0;
p1 = n1[lambda / 1000] * Cos[θ];
k1 = (2 * Pi / lambda) * n1[lambda / 1000] * Cos[θ];
m1 =
  {{Cos[k1 * h1], (-I / p1) * Sin[k1 * h1]}, {-I * p1 * Sin[k1 * h1], Cos[k1 * h1]}};
p2 = n2[lambda / 1000] * Cos[θ];
k2 = (2 * Pi / lambda) * n2[lambda / 1000] * Cos[θ];
m2 =
  {{Cos[k2 * h2], (-I / p2) * Sin[k2 * h2]}, {-I * p2 * Sin[k2 * h2], Cos[k2 * h2]}};
pd = nd * Cos[θ];
kd = (2 * Pi / lambda) * nd * Cos[θ];
md =
  {{Cos[kd * hd], (-I / pd) * Sin[kd * hd]}, {-I * pd * Sin[kd * hd], Cos[kd * hd]}};
p0 = 1; pf = 1;
m3 = MatrixPower[m1.m2, Nbef].md.MatrixPower[m2.m1, Naft];
t = (2 * p0) / ((m3[[1, 1]] + pf * m3[[1, 2]]) * p0 + (m3[[2, 1]] + pf * m3[[2, 2]]));
r = ((m3[[1, 1]] + pf * m3[[1, 2]]) * p0 - (m3[[2, 1]] + pf * m3[[2, 2]])) /
   ((m3[[1, 1]] + pf * m3[[1, 2]]) * p0 + (m3[[2, 1]] + pf * m3[[2, 2]]));
T = (pf / p0) * Abs[t^2];
R = Abs[r^2];
Plot[T /. nd → 1.33, {lambda, 430, 1530}, PlotRange → All]
Plot[R /. nd → 1.33, {lambda, 430, 1530}, PlotRange → All]
```






```
(*1st zone*)
Max5 = FindMaximum[{T /. nd → 1.33, 790 ≤ lambda ≤ 810}, {lambda, 800}]
Max6 = FindMaximum[{T /. nd → 1.331, 790 ≤ lambda ≤ 810}, {lambda, 800}]
(*sensitivity*)
(Abs[Max5[[2, 1, 2]] - Max6[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Max5[[2, 1, 2]] - Max6[[2, 1, 2]]]) / (0.001 * 803.6189610364894)
```

{1., {lambda → 803.619}}

{1., {lambda → 803.872}}

253.118

0.314972

```
(*1st zone*)
Min5 = FindMinimum[{R /. nd → 1.33, 790 ≤ lambda ≤ 810}, {lambda, 800}]
Min6 = FindMinimum[{R /. nd → 1.331, 790 ≤ lambda ≤ 810}, {lambda, 800}]
(*sensitivity*)
(Abs[Min5[[2, 1, 2]] - Min6[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Min5[[2, 1, 2]] - Min6[[2, 1, 2]]]) / (0.001 * 803.6189610364414)
```

$\{1.60928 \times 10^{-10}$, {lambda → 803.619}$\}$

$\{1.74368 \times 10^{-10}$, {lambda → 803.872}$\}$

253.118

0.314972